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ACME PETROCHEMICAL DEMO COMPLEX - UNIT 2300

PROCESS DATA SHEET: SHELL & TUBE HEAT EXCHANGER

Document Code: DS-E2307 | Revision: Z1 | Status: Issued for Construction

Equipment Tag: E-2307

Equipment Name: Decomposer Reactor Circulation Cooler

Equipment Class: Shell and Tube Heat Exchanger (TEMA Type BEM)

Process Unit: Unit 2300 (Decomposition & Recovery Section)

Service: Removes exothermic heat of CHP cleavage (-4.85 Gcal/hr) from D-2304 recirculation loop

1. THERMAL & HYDRAULIC DESIGN DATA

- Heat Duty: 4.85 Gcal/hr (5,640 kW)
- Effective Heat Transfer Area: 312.0 m²
- TEMA Designation: BEM (One-Pass Shell, Two-Pass Fixed Tubesheet)
- Tube Side (Process Fluid - Stream S379 -> S380):
 - * Fluid: Phenol / Acetone / Cumene / Residual CHP Reaction Mixture
 - * Mass Flow Rate: 185,000 kg/hr
 - * Inlet / Outlet Temperature: 82.0 degC / 74.0 degC
 - * Design Pressure / Temperature: 11.0 kg/cm²g / 250 degC
 - * Material: SA-213 Type 316L Seamless Tubes (OD 19.05 mm x 2.11 mm BWG, Length 6096 mm)
- Shell Side (Cooling Medium - Tempered Cooling Water):
 - * Mass Flow Rate: 485,000 kg/hr modulated by Control Valve TV-1302
 - * Inlet / Outlet Temperature: 32.0 degC / 42.0 degC
 - * Design Pressure / Temperature: 7.0 kg/cm²g / 100 degC
 - * Material: Carbon Steel SA-516 Gr.70

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DS-E2307 PAGE 2 OF 4 - MECHANICAL CONSTRUCTION & BAFFLE GEOMETRY

Equipment Tag: E-2307

- Shell ID: 1050 mm | Tube Count: 842 Tubes on 23.8 mm Triangular Pitch
- Baffle Type: Single Segmental (25% Cut), Baffle Spacing 350 mm
- Tube-to-Tubesheet Joint: Strength Welded + Hydraulically Expanded (Lethal / Peroxide Service)

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DS-E2307 PAGE 3 OF 4 - CONNECTIONS & INSTRUMENTATION

Equipment Tag: E-2307

- Process Inlet Nozzle T1 (10 in 300#): Receives Stream S379 from D-2304 via P-2304A/B
- Process Outlet Nozzle T2 (10 in 300#): Returns cooled Stream S380 to D-2304
- Cooling Water Outlet Valve TV-1302: Modulated by D-2304 temperature controller TIC-1302 (fails 100% open on air)

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DS-E2307 PAGE 4 OF 4 - INSPECTION & HYDROTEST REQUIREMENTS

Equipment Tag: E-2307

- Tube Side Hydrotest Pressure: 16.5 kg/cm²g with Chloride-Free Demineralized Water (< 1 ppm Cl⁻)
- Shell Side Hydrotest Pressure: 10.5 kg/cm²g